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absence of strong centralized state systems, feudal relationships held societies together.

From the 11th century, a revival of Western European culture, trade, and urban life gathered pace. The “Medieval Warm Period” (950–1250), when Europe experienced above-average temperatures, improved yields from agriculture; it was also a time when great cathedrals and castles were constructed. But even when the Christian crusaders fought their way to Jerusalem at the heart of the Muslim world, the flow of civilization was the other way, with Islamic scholars far advanced in medicine, philosophy, astronomy, and geography.

Expansion and contraction

By the 13th century, the world's population is believed to have risen to around 400 million—double its total at the high point of the ancient empires. A wide-ranging network linked Europe to China and the thriving trading kingdoms of Asia, by land along the Silk Road and by sea across the Indian Ocean. Cairo and Venice both became wealthy cities as focal points at the western end of this trade.

However, civilized life remained precarious. The Mongols—nomadic warriors from the Asian steppes—seized major cities

from the Middle East to southern China, carrying out large-scale massacres. Lethal diseases were also highly prevalent. Carried along the trade routes in the mid-14th century, the Black Death epidemic may have killed a quarter of the world's population.

Inventions and progress

Technological progress was slow but cumulatively substantial. As the world's most advanced country, China was the ultimate source of most inventions, from paper and printing to the magnetic compass and gunpowder. Even relatively backward Europe benefited from improvements in shipbuilding and metalworking, and the invention and spread of the plow and the windmill transformed agriculture.

By the end of the Middle Ages, Western European kingdoms had developed from “feudal” states, based on oaths of loyalty, to more stable and centralized states, able to channel their key resources into the large projects of colonization and exploration. In the Americas, meanwhile, civilizations such as the Aztecs and Incas continued to evolve independently, untouched by developments in Eurasia and Africa, until the Spanish conquistadors arrived in the 16th century. ■